UCS3461 Foundations of Artificial Intelligence

BAYES RULE

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Model and Inference

- Model by a probability distribution $P(X_1, X_2, \ldots, X_n)$
  - Choose random variables
  - Find conditional independencies
  - For each random variable, Conditional Probability Table (CPT)
- From $P(X_1, X_2, \ldots, X_n)$ derive $P(X_i)$ or $P(X_i|X_j, \ldots, X_k)$

Sum Rule, Product Rule, Bayes Rule

Sum Rule, Product Rule, Bayes Rule

- Sum rule

$$P(X) = \sum_Y P(X, Y)$$

- Product rule

$$P(X, Y) = P(X)P(Y|X)$$

$$P(Y|X) = \frac{P(X, Y)}{P(X)}$$

- Bayes rule

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

Conditional Independence

X and Y are conditionally independent on Z

$$P(X, Y|Z) = P(X|Y, Z)P(Y|Z) = P(X|Z)P(Y|Z)$$
$$P(Y|X, Z) = P(Y|Z)$$

Conditional Independence: Example

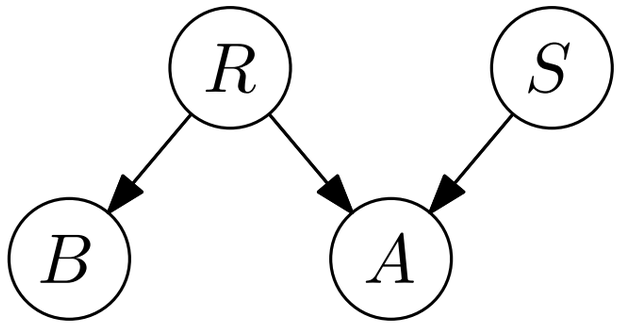
One morning Anbu leaves her house and realises that her grass is wet. Is it due to overnight rain or did she forget to turn off the sprinkler last night? Next she notices that the grass of her neighbour, Babu, is also wet. This explains away to some extent the possibility that her sprinkler was left on, and she concludes therefore that it has probably been raining.

Conditional Independence: Example

P(r)
0.2

P(s)
0.1

R	P(b|R)
0	0.2
1	1



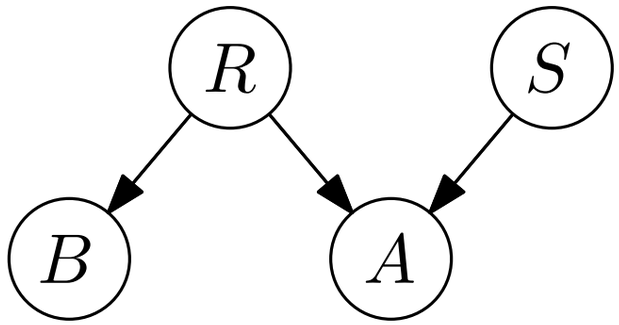
R	S	P(a|R,S)
0	0	0
0	1	0.9
1	0	1
1	1	1

Conditional Independence: Example

R	P(R)
0	0.8
1	0.2

S	P(S)
0	0.9
1	0.1

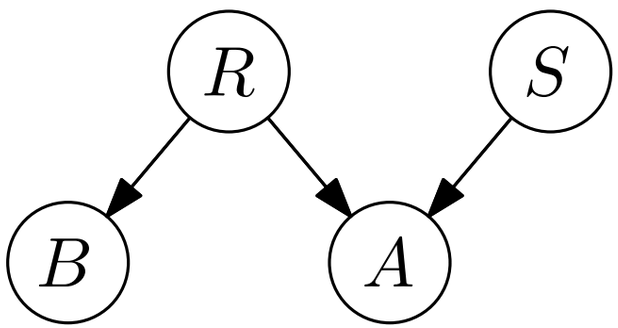
B	R	P(B|R)
0	0	0.8
0	1	0
1	0	0.2
1	1	1



A	R	S	P(A|R,S)
0	0	0	1
0	0	1	0.1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0.9
1	1	0	1
1	1	1	1

Conditional Independence

- Without conditional independence
  - $P(A, B, R, S) = P(A|B, R, S)P(B|R, S)P(R|S)P(S)$
  - $2^3 + 2^2 + 2^1 + 2^0 = 8 + 4 + 2 + 1 = 15$
- With conditional independence
  - $P(A, B, R, S) = P(A|R, S)P(B|R)P(R)P(S)$
  - $2^2 + 2^1 + 1 + 1 = 8$



Inference

$$
\begin{aligned}
&P(S=1|A=1,B=1)\\
&= \frac{P(S=1,A=1,B=1)}{P(A=1,B=1)}\\
&\propto P(S=1,A=1,B=1)\\
&= \sum_R P(A=1,B=1,R,S=1)\\
&= \sum_R P(A=1|R,S=1)P(B=1|R)P(R)P(S=1)\\
&= \quad P(A=1|R=0,S=1)P(B=1|R=0)P(R=0)P(S=1)\\
&\quad + P(A=1|R=1,S=1)P(B=1|R=1)P(R=1)P(S=1)\\
&= \quad 0.9\times 0.2\times 0.8\times 0.1 + 1\times 1\times 0.2\times 0.1\\
&= 0.0344\\
&P(S=1|A=1,B=1) \propto 0.0344
\end{aligned}
$$

B	R	P(B|R)
0	0	0.8
0	1	0
1	0	0.2
1	1	1

A	R	S	P(A|R,S)
0	0	0	1
0	0	1	0.1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0.9
1	1	0	1
1	1	1	1

P(r)	P(s)
0.2	0.1

Inference

$$
\begin{aligned}
&P(S=0|A=1,B=1)\\
&= \frac{P(S=0,A=1,B=1)}{P(A=1,B=1)}\\
&\propto P(S=0,A=1,B=1)\\
&= \sum_R P(A=1,B=1,R,S=0)\\
&= \sum_R P(A=1|R,S=0)P(B=1|R)P(R)P(S=0)\\
&= \quad P(A=1|R=0,S=0)P(B=1|R=0)P(R=0)P(S=0)\\
&\quad + P(A=1|R=1,S=0)P(B=1|R=1)P(R=1)P(S=0)\\
&= \quad 0\times\ldots+1\times1\times0.2\times0.9\\
&= 0.18\\
&P(S=0|A=1,B=1)\propto 0.18
\end{aligned}
$$

B	R	P(B|R)
0	0	0.8
0	1	0
1	0	0.2
1	1	1

A	R	S	P(A|R,S)
0	0	0	1
0	0	1	0.1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0.9
1	1	0	1
1	1	1	1

P(r)	P(s)
0.2	0.1

Inference

$$
\begin{aligned}
&P(S=1|A=1,B=1) \\
&= \frac{P(S=1,A=1,B=1)}{P(S=1,A=1,B=1)+P(S=0,A=1,B=1)} \\
&= \frac{0.0344}{0.0344+0.18} \\
&= 0.1604
\end{aligned}
$$

Inference

$$
\begin{aligned}
&P(S=1|A=1)\\
&= \frac{P(S=1, A=1)}{P(A=1)}\\
&= \frac{\sum_{B,R} P(A=1, B, R, S=1)}{\sum_{B,R,S} P(A=1, B, R, S)}\\
&= \frac{\sum_{B,R} P(A=1|R, S=1)P(B|R)P(R)P(S=1)}{\sum_{B,R,S} P(A=1|R, S)P(B|R)P(R)P(S)}\\
&= \frac{\sum_{R} P(A=1|R, S=1)P(R)P(S=1)}{\sum_{R,S} P(A=1|R, S)P(R)P(S)}\\
&= \frac{0.9 \times 0.8 \times 0.1 + 1 \times 0.2 \times 0.1}{0.9 \times 0.8 \times 0.1 + 1 \times 0.2 \times 0.1 + 0 \times 0.8 \times 0.9 + 1 \times 0.2 \times 0.9}\\
&= 0.3382
\end{aligned}
$$

B	R	P(B|R)
0	0	0.8
0	1	0
1	0	0.2
1	1	1

A	R	S	P(A|R,S)
0	0	0	1
0	0	1	0.1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0.9
1	1	0	1
1	1	1	1

P(r)	P(s)
0.2	0.1

Bayes Rule

Bayes Rule

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

Posterior Likelihood Prior

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}$$

Evidence

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

From Causal to Diagnostic

- From causal direction to diagnostic direction

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

$$P(\text{disease}|\text{symptom}) = \frac{P(\text{symptom}|\text{disease})P(\text{disease})}{P(\text{symptom})}$$

Bayes Rule: Example

Suppose $P(S) = \frac{1}{2}$

$$P(D) = \frac{1}{100,000}$$

$$P(S|D) = 0.9$$

$$P(D|S) = \frac{P(S|D)P(D)}{P(S)}$$

$$= \frac{0.9 \times \frac{1}{100,000}}{\frac{1}{2}}$$

$$= 1.8 \times 10^{-5}$$

Suppose $P(S) = \frac{1}{1000}$

$$P(D) = \frac{1}{100,000}$$

$$P(S|D) = 0.9$$

$$P(D|S) = \frac{P(S|D)P(D)}{P(S)}$$

$$= \frac{0.9 \times \frac{1}{100,000}}{\frac{1}{1000}}$$

$$= 0.009$$

Let D and S be random variables:

D Disease

S Symptom

Bayes Rule: Application

$$
\begin{aligned}
P(y|\mathbf{x}) &= \frac{P(\mathbf{x}|y)P(y)}{P(\mathbf{x})} \\
&= \frac{P(\mathbf{x}|y)P(y)}{\sum_y P(\mathbf{x}, y)} \\
&= \frac{P(\mathbf{x}|y)P(y)}{\sum_y P(\mathbf{x}|y)P(y)} \\
&= \frac{P(\mathbf{x}|y)P(y)}{P(\mathbf{x}|y=0)P(y=0) + P(\mathbf{x}|y=1)P(y=1)}
\end{aligned}
$$

Bayes Rule: Application

$$
\begin{aligned}
P(y|\mathbf{x}) &= \frac{P(\mathbf{x}|y)P(y)}{P(\mathbf{x})} \\
&\propto P(\mathbf{x}|y)P(y) \\
P(\neg y|\mathbf{x}) &= \frac{P(\mathbf{x}|\neg y)P(\neg y)}{P(\mathbf{x})} \\
&\propto P(\mathbf{x}|\neg y)P(\neg y) \\
P(y|\mathbf{x}) &= \frac{P(y|\mathbf{x})}{P(y|\mathbf{x}) + P(\neg y|\mathbf{x})}
\end{aligned}
$$

Naive Bayes and Conditional Independence

Naive Bayes and Conditional Independence

$$
\begin{aligned}
P(y|\mathbf{x}) &= \frac{P(\mathbf{x}|y)P(y)}{\sum_y P(\mathbf{x}|y)P(y)} \\
&= \frac{P(x_1, \ldots, x_n|y)P(y)}{\sum_y P(x_1, \ldots, x_n|y)P(y)} \\
&= \frac{P(x_1|y)P(x_2|y)\ldots P(x_n|y)P(y)}{\sum_y P(x_1|y)P(x_2|y)\ldots P(x_n|y)P(y)}
\end{aligned}
$$

Naive Bayes assumption

$$P(x_1, x_2, \ldots, x_n|y) = P(x_1|y)P(x_2|y)\ldots P(x_n|y)$$

Estimation Using Maximum Likelihood

- Dataset $= \{(\mathbf{x}^n, y^n), n = 1, \ldots, N\}$
- Binary attributes: $x_i^n \in \{0, 1\}, i = 1, \ldots, D$
- Binary classification: $y = 0$ or $y = 1$

Likelihood Function

- Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with a parameter y. Suppose that we have observed $X_1 = x_1, X_2 = x_2, \ldots, X_n$.
- If X_i's are discrete, then the likelihood function is defined as

$$L(x_1, x_2, \ldots, x_n; y) = P_{X_1 X_2 \ldots X_n}(x_1, x_2, \ldots, x_n | y)$$

In some problems, it is easier to work with the log likelihood function given by

$$\ln L(x_1, x_2, \ldots, x_n | y)$$

Example

	x_0	x_1	x_2	x_3	x_4	y
1	0	0	1	1	1	0
2	1	0	1	1	0	0
3	1	1	0	0	1	0
4	1	1	0	0	0	0
5	0	1	0	0	1	0
6	0	0	0	1	0	0
1	1	0	0	1	1	1
2	1	1	0	0	1	1
3	1	1	1	1	0	1
4	1	1	0	1	0	1
5	1	1	0	1	1	1
6	1	0	1	1	0	1
7	1	0	1	0	0	1

- Classify $\mathbf{x} = [1, 0, 1, 1, 0]^T \equiv$
- What is posterior probability $P(y|\mathbf{x}) = P(y|x_1, \neg x_2, x_3, x_4, \neg x_5)$?
- Bayes' rule and Naive Bayes assumption
  $P(y|\mathbf{x}) \propto P(\mathbf{x}|y)P(y)$
  $P(\neg y|\mathbf{x}) \propto P(\mathbf{x}|\neg y)P(y)$
- Prior probability:
  $P(\neg y) = 6/13, P(y) = 7/13$

Naive Bayes Assumption

- $P(y|\mathbf{x}) = \dfrac{P(\mathbf{x}|y)P(y)}{P(\mathbf{x}|\neg y)P(\neg y) + P(\mathbf{x}|y)P(y)}$
- Naive Bayes assumption
  $P(\mathbf{x}|y) = P(x_1, x_2, x_3, x_4, x_5|y) =$
  $P(x_1|y)P(x_1|y)P(x_2|y)P(x_3|y)P(x_4|y)P(x_5|y)$
- $P(\mathbf{x} = [1, 0, 1, 1, 0]^T|y = 1)$
  $P(x_1|y)P(\neg x_2|y)P(x_3|y)P(x_4|y)P(\neg x_5|y)$
- $P(\mathbf{x} = [1, 0, 1, 1, 0]^T|y = 0)$
  $P(x_1|\neg y)P(\neg x_2|\neg y)P(x_3|\neg y)P(x_4|\neg y)P(\neg x_5|\neg y)$

Likelihood Calculation

	x_0	x_1	x_2	x_3	x_4	y
1	0	0	1	1	1	0
2	1	0	1	1	0	0
3	1	1	0	0	1	0
4	1	1	0	0	0	0
5	0	1	0	0	1	0
6	0	0	0	1	0	0
1	1	0	0	1	1	1
2	1	1	0	0	1	1
3	1	1	1	1	0	1
4	1	1	0	1	0	1
5	1	1	0	1	1	1
6	1	0	1	1	0	1
7	1	0	1	0	0	1

$$P(x_1|\neg y) = 3/6 = 1/2$$
$$P(x_2|\neg y) = 3/6 = 1/2$$
$$P(x_3|\neg y) = 2/3 = 1/3$$
$$P(x_4|\neg y) = 3/6 = 1/2$$
$$P(x_5|\neg y) = 3/6 = 1/2$$
$$P(x_1|y) = 7/7 = 1$$
$$P(x_2|y) = 4/7$$
$$P(x_3|y) = 3/7$$
$$P(x_4|y) = 5/7$$
$$P(x_5|y) = 3/7$$

Likelihood

$$P(x_1 = 1|y = 0) = 1/2 \qquad P(x_1 = 1|y = 1) = 1$$
$$P(x_2 = 1|y = 0) = 1/2 \qquad P(x_2 = 1|y = 1) = 4/7$$
$$P(x_3 = 1|y = 0) = 1/3 \qquad P(x_3 = 1|y = 1) = 3/7$$
$$P(x_4 = 1|y = 0) = 1/2 \qquad P(x_4 = 1|y = 1) = 5/7$$
$$P(x_5 = 1|y = 0) = 1/2 \qquad P(x_5 = 1|y = 1) = 3/7$$

$$\begin{aligned}
P(\mathbf{x} = [1,0,1,1,0]^T|y = 1) \\
&= P(x_1 = 1|y = 1)P(x_2 = 0|y = 1)P(x_3 = 1|y = 1) \\
&\quad P(x_4 = 1|y = 1)P(x_5 = 0|y = 1) \\
&= 1 \times \frac{3}{7} \times \frac{3}{7} \times \frac{5}{7} \times \frac{4}{7} = 0.075 \\
P(\mathbf{x} = [1,0,1,1,0]^T|y = 0) \\
&= P(x_1 = 1|y = 0)P(x_2 = 0|y = 0)P(x_3 = 1|y = 0) \\
&\quad P(x_4 = 1|y = 0)P(x_5 = 0|y = 0) \\
&= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} \times \frac{1}{2} = 0.021
\end{aligned}$$

Posterior Probability

- $P(y=0) = 6/13, P(y=1) = 7/13$

$$
\begin{aligned}
& P(y=1|\mathbf{x}) \\
=& \frac{P(\mathbf{x}|y=1) \quad P(y=1)}{P(\mathbf{x}|y=0)P(y=0) + P(\mathbf{x}|y=1)P(y=1)} \\
=& \frac{1 \times \frac{3}{7} \times \frac{3}{7} \times \frac{5}{7} \times \frac{4}{7} \times \frac{7}{13}}{1 \times \frac{3}{7} \times \frac{3}{7} \times \frac{5}{7} \times \frac{4}{7} \times \frac{7}{13} + \frac{1}{2} \times \frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} \times \frac{1}{2} \times \frac{6}{13}} \\
=& 0.8076
\end{aligned}
$$

Posterior Odds

- $P(y=1|\mathbf{x}) = \dfrac{P(\mathbf{x}|y=1)P(y=1)}{P(\mathbf{x})}$
- $\dfrac{P(y=1|\mathbf{x})}{P(y=0|\mathbf{x})} = \dfrac{P(\mathbf{x}|y=1)P(y=1)}{P(\mathbf{x}|y=0)P(y=0)}$
- $\dfrac{P(y=1|\mathbf{x}=[1,0,1,1,0]^T)}{P(y=0|\mathbf{x}=[1,0,1,1,0]^T} = \dfrac{P(\mathbf{x}=[1,0,1,1,0]^T)|y=1)P(y=1)}{P(\mathbf{x}=[1,0,1,1,0]^T|y=0)P(y=0)}$

  $\dfrac{P(x_1=1|y=1)P(x_2=0|y=1)P(x_3=1|y=1)P(x_4=1|y=1)P(x_5=0|y=1)P(y}{P(x_1=1|y=0)P(x_2=0|y=0)P(x_3=1|y=0)P(x_4=1|y=0)P(x_5=0|y=0)P(y}$
- $\dfrac{1 \times \frac{3}{7} \times \frac{3}{7} \times \frac{5}{7} \times \frac{4}{7} \times \frac{7}{13}}{\frac{1}{2} \times \frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} \times \frac{1}{2} \times \frac{6}{13}}$
- 4.17

Likelihood Ratio

- $P(y=1|\mathbf{x}) = \dfrac{P(\mathbf{x}|y=1)P(y=1)}{P(\mathbf{x})}, P(y=0|\mathbf{x}) = \dfrac{P(\mathbf{x}|y=0)P(y=0)}{P(\mathbf{x})}$
- $\dfrac{P(y=1|\mathbf{x})}{P(y=0|\mathbf{x})} = \dfrac{P(\mathbf{x}|y=1)P(y=1)}{P(\mathbf{x}|y=0)P(y=0)} = \dfrac{P(\mathbf{x}|y=1)}{P(\mathbf{x}|y=0)}$
- $\dfrac{P(y=1|\mathbf{x}=[1,0,1,1,0]^T)}{P(y=0|\mathbf{x}=[1,0,1,1,0]^T} = \dfrac{P(\mathbf{x}=[1,0,1,1,0]^T)|y=1)}{P(\mathbf{x}=[1,0,1,1,0]^T|y=0)}$

$$\frac{P(x_1=1|y=1)P(x_2=0|y=1)P(x_3=1|y=1)P(x_4=1|y=1)P(x_5=0|y=}{P(x_1=1|y=0)P(x_2=0|y=0)P(x_3=1|y=0)P(x_4=1|y=0)P(x_5=0|y=}$$

- $\dfrac{1 \times \frac{3}{7} \times \frac{3}{7} \times \frac{5}{7} \times \frac{4}{7}}{\frac{1}{2} \times \frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} \times \frac{1}{2}}$
- 3.57

Exercises

A surveying company partitions radio station listeners into two groups – the `young` and `old`. They assume that, given the knowledge that a customer is either `young` or `old`, this is sufficient to determine whether or not a customer will like a particular radio station, independent of their likes or dislikes for any other stations:

$$p(r1, r2, r3, r4|age) = p(r1|age)p(r2|age)p(r3|age)p(r4|age)$$

where each of the variables $r1, r2, r3, r4$ can take the states `like` or `dislike`, and the `age` variable can take the value `young` or `old`. Thus the information about the age of the customer determines the individual radio station preferences without needing to know anything else. To complete the specification, given that a customer is young, she has a 95% chance to like Radio1, a 5% chance to like Radio2, a 2% chance to like Radio3 and a 20% chance to like Radio4. Similarly, an old listener has a 3% chance to like Radio1, an 82% chance to like Radio2, a 34% chance to like Radio3 and a 92% chance to like Radio4. They know that 90% of the listeners are old. Given this model, and the fact that a new customer likes Radio1, and Radio3, but dislikes Radio2 and Radio4, what is the probability that the new customer is `young`?

Exercises

A psychologist does a small survey on 'happiness'. Each respondent provides a vector with entries 1 or 0 corresponding to whether they answer 'yes' to a question or 'no', respectively. The question vector has attributes: $\mathbf{x}$ =(rich, married, healthy). Thus, a response (1, 0, 1) would indicate that the respondent was 'rich', 'unmarried', 'healthy'. In addition, each respondent gives a value c = 1 if they are content with their lifestyle, and c = 0 if they are not. The following responses were obtained from people who claimed also to be 'content' : (1, 1, 1), (0, 0, 1), (1, 1, 0), (1, 0, 1) and for 'not content': (0, 0, 0), (1, 0, 0), (0, 0, 1), (0, 1, 0), (0, 0, 0). Using Naive Bayes, calculate the probability that a person who is 'not rich' and 'married' is 'content' ? (That is, we do not know whether or not they are 'healthy'). Show the steps of the calculation.

Summary

- Rules of probability: sum rule and product rule
- Bayes theorem
- Inference using Bayes theorem

End